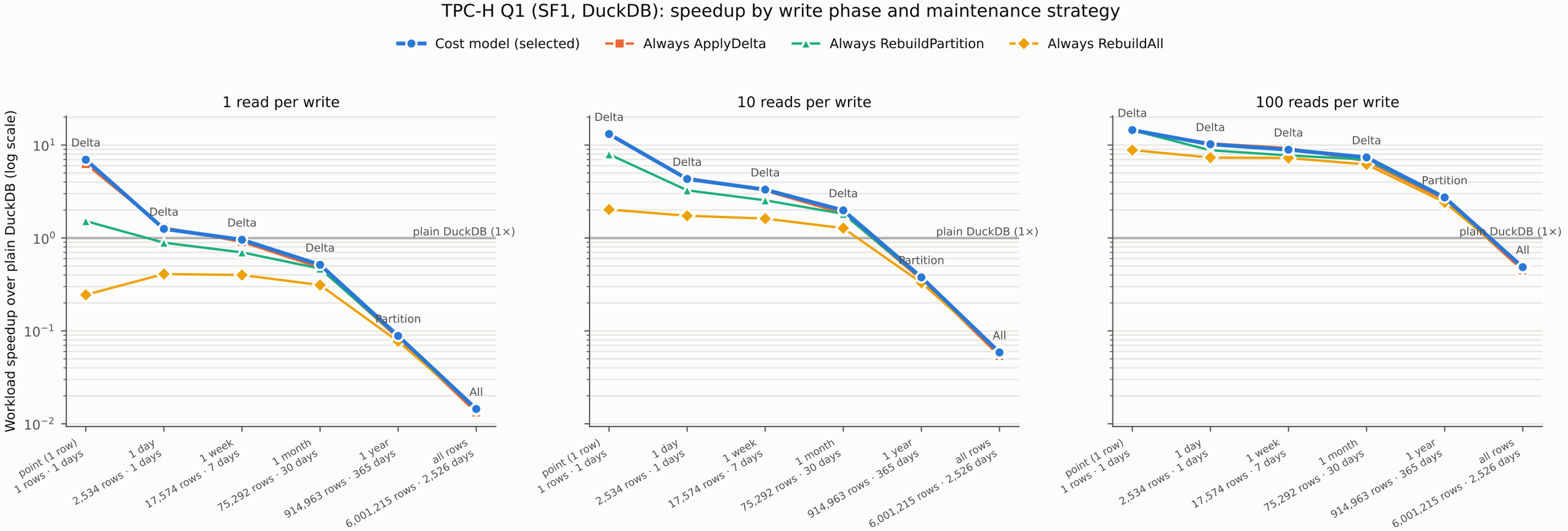

Per phase: 4 writes (UPDATE l_quantity on the phase's rows), each followed by reads of Q1. Cost per write = write time (incl. capture and maintenance) + reads × read time. Labels on the cost-model line: the action it chose.

Measurements (mean per write and per read, SF1; median of the runs)

Phase	Rows	Days	Plain DuckDB write / read ms	Cost model (selected write / read ms	Always ApplyDelta write / read ms	Always RebuildPartitio write / read ms	Always RebuildAll write / read ms	Chosen action
point (1 row)	1	1	1.4 / 23.3	2.0 / 1.58	2.4 / 1.59	14.8 / 1.46	99.2 / 1.64	ApplyDelta
1 day	2,534	1	32.2 / 23.2	42.3 / 1.88	40.9 / 1.84	60.2 / 2.07	133.4 / 1.86	ApplyDelta
1 week	17,574	7	36.7 / 23.0	60.3 / 2.02	64.0 / 1.88	83.1 / 2.19	147.6 / 1.75	ApplyDelta
1 month	75,292	30	36.7 / 22.9	113.9 / 2.02	122.2 / 2.25	125.2 / 2.07	189.6 / 1.86	ApplyDelta
1 year	914,963	365	38.8 / 23.1	699.8 / 1.61	741.7 / 1.69	717.0 / 1.60	801.6 / 1.71	RebuildPartition
all rows	6,001,215	2,526	44.9 / 23.3	4719.8 / 1.69	5143.8 / 1.59	4728.7 / 1.69	4661.4 / 1.81	RebuildAll